PW AITS
ALL INDIA TEST SERIES

11th JEE Main

AJAM7/06 **Test- 06**

DURATION : 180 Minutes **DATE : 23/02/2025** **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus- 11th
Chemistry:	Full Syllabus- 11th
Mathematics:	Full Syllabus- 11th

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

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SECTION-I (PHYSICS)

Single Correct Type Questions

1. Two identical celestial bodies of masses m are floating in space, exerting a gravitational pull of magnitude F on each other. A portion of mass (Δm) is transferred from one body to the other, and the distance between them is reduced to two-thirds of their initial separation. As a result, the gravitational force between them changes to $\frac{8}{9}$ of the original force. Determine the ratio of mass transferred (Δm) from one body to the other to the original mass m i.e. $\left(\frac{\Delta m}{m}\right)$.

(1) $\frac{1}{3}$ (2) $\frac{7}{9}$

(3) $\frac{4}{5}$ (4) $\frac{2}{3}$

2. A heavy rope with a uniform mass distribution, total mass M, and length L is dropped over the edge of a cliff. Part of the rope hangs down the cliff, while the rest lies flat on horizontal top. A person with mass $\frac{M}{6}$ is clinging to the very end of the hanging portion. A group of friends exerts a steady force to pull both the hanging rope and the person up to the top of the cliff. For their efforts, the minimum required work done by the group is calculated to be $\frac{MgL}{9}$. What is the length of the portion of the rope that was hanging off the cliff?

(1) $\frac{L}{3}$ (2) $\frac{L}{9}$

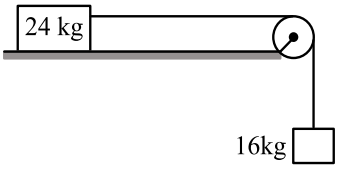
(3) $\frac{L}{18}$ (4) $\frac{4L}{9}$

3. A thermally insulated box is divided by a partition into two compartments, each having volume V. Initially one compartment contains 1 mole of a monoatomic ideal gas at temperature 298 K, and the other is evacuated. The partition is then broken, and the gas expands to fill both compartments. Its temperature change is :

(1) zero (2) 7.76 K

(3) 8.86 K (4) 10.86 K

4. A professor conducted an experiment to investigate the dynamics of a connected system on a rough horizontal surface. An object of mass 24 kg is placed on a rough plane with a coefficient of kinetic friction of 0.5. The object is connected via an inextensible, massless string to a point mass 16 kg suspended over a frictionless pulley. Assuming the gravitational acceleration is 10ms^{-2}, the force experienced by the clamp on the pulley is close to



(1) 75 N (2) 252 N

(3) 204 N (4) 116 N

5. A system composed of a collection of particles forming a droplet of radius r exhibits a pressure described by the following equation $P = \alpha e^{-\frac{2\beta V}{nrRT}}$, where P represents the pressure within the droplet, α and β are constants characterizing the properties of the droplet, V is the Volume, R is the gas constant, n is the number of moles, T is the absolute temperature of the droplet. Determine the dimensions of the constant $\alpha \times \beta$.

(1) $[M^0 L^{-1} T^0]$ (2) $[M^2 L^{-1} T^{-4}]$

(3) $[M^0 L^1 T^0]$ (4) $[M\,LT^{-2}]$

6. A thin wire of length x, connected to a stationary point, stretches under the weight of suspended masses. When a 10-kilogram mass is attached, the wire elongates to a length x_1. If the mass is doubled to 20 kilograms, the wire's length extends further to x_2. Calculate the stress experienced by the wire due to the 10-kilogram mass, considering Young's modulus, Y, as a measure of the wire's elasticity.

(1) $\left(\frac{x_2 - x_1}{2x_1 - x_2}\right)Y$ (2) $\left(\frac{x_1 + x_2}{2x_1 - x_2}\right)Y$

(3) $\left(\frac{x_2 + x_1}{2x_1 + x_2}\right)Y$ (4) $\left(\frac{x_2 - x_1}{2x_1 + x_2}\right)Y$

7. A young physicist stands on the edge of a 11.25 meter high platform and projects a stone at some angle with horizontal into the air. Upon descending, the stone impacts the ground with a final velocity of 25 m/s. Assuming air resistance is negligible, determine the initial velocity at which the stone was launched. (take $g = 10\text{m/s}^2$).

(1) 20 m/s (2) 15 m/s

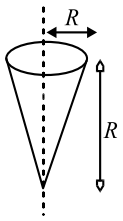
(3) 10 m/s (4) $10\sqrt{5}$ m/s

8. A vehicle travels on a rough banked curve with a radius of 10 meters and a banking angle of 45°. The maximum permissible velocity, beyond which the vehicle begins to skid, is determined to be 20 m/s. Assuming uniform circular motion and neglecting air resistance, the coefficient of static friction between the tires and the road surface is (gravitational acceleration is given as $g = 10\ \text{m/s}^2$).

(1) 0.17 (2) 0.30

(3) 0.60 (4) 0.45

9. Consider a solid cone of mass *M*, base radius *R*, and height *R* as depicted in the figure. The cone is symmetrically bisected by a plane perpendicular to its axis at the midpoint of its height, producing a truncated conical frustum. The moment of inertia of this frustum about its central axis of symmetry is



(1) $\frac{31MR^2}{64}$ (2) $\frac{7MR^2}{16}$

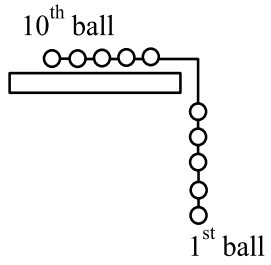
(3) $\frac{9MR^2}{40}$ (4) $\frac{93MR^2}{320}$

10. A sealed adiabatic cylinder of length *l* is divided into two chambers by a thin, frictionless, thermally insulated piston of mass *m*. The cylinder is placed vertically such that the piston can move freely along the cylinder's length without any heat transfer. The ratio of length of the upper chamber, and the lower chamber is 3:2. Both chambers contain 3 moles of monoatomic ideal gas at the same initial temperature *T*. If the piston is displaced slightly from equilibrium, it undergoes oscillatory motion. The time period of the oscillation, is $\left(\text{Take } \ell\sqrt{\frac{m}{RT}} = \frac{10}{\pi}\sqrt{65}\right)$

(1) 24 sec (2) 16 sec

(3) 32 sec (4) 8 sec

11. Consider a system comprising ten spherically shaped objects, each possessing a mass of 2 kilograms. These objects are interconnected by an inextensible string of negligible mass. The system is positioned such that a portion of it extends over the edge of a frictionless horizontal surface, as depicted in the diagram. Determine the tension (in N) acting upon the string segment connecting the sixth and seventh objects at the instant when the fifth object precisely detaches from the surface. The acceleration due to gravity is given as $10\ \text{m/s}^2$.



(1) 30 (2) 40

(3) 50 (4) 60

12. Given two vectors, $\vec{A}$ and $\vec{B}$, that satisfy the following vector equations $\vec{A}\cdot\vec{B} = \left|\vec{A}\times\vec{B}\right|$ and $\left(\vec{A}+\vec{B}\right)\cdot\left(\vec{A}-\vec{B}\right) = 0$. Determine the magnitude of the vector sum $\left|\vec{A}+\vec{B}\right|$.

(1) $\sqrt{2-\sqrt{2}}A$

(2) $\sqrt{2}A$

(3) $2A$

(4) $\sqrt{2+\sqrt{2}}A$

13. A sonometer wire of length 1.5 m is made of steel. The tension in it produces an longitudinal elastic strain of 1%. If density and Young's modulus of steel are $7.7 \times 10^3\ \text{kg/m}^3$ and $2.2 \times 10^{11}\ \text{N/m}^2$, the fundamental frequency of the wire (in Hz) is nearly

(1) 178 (2) 300

(3) 700 (4) 288

14. In a precision measurement experiment utilizing Vernier callipers with a resolution of 0.05 mm, the instrument's zero error must be accounted for. It was observed that when the calliper jaws are brought into contact, the zero mark of the vernier scale aligns right to the zero mark of the main scale. Additionally, the eighth division of the vernier scale coincides with a main scale division. During the measurement of a spherical object's diameter, the zero of the Vernier scale falls right to mark of 4.20 cm, and the sixth Vernier scale division coincides exactly with a main scale division. What is the corrected diameter of the spherical object? (Consider 19 MSD = 20VSD)

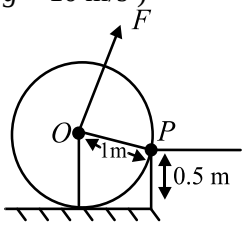
(1) 4.19 cm

(2) 4.27 cm

(3) 4.17 cm

(4) 4.22 cm

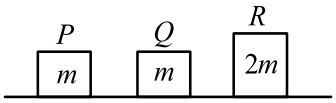
15. A cylindrical body of homogeneous mass distribution, weighing 5 kilograms and possessing a radius of 1 meter, is to be surmounted over a step of height 0.5 meters. To accomplish this, a force F is applied at its center of mass, oriented perpendicularly to the plane that contains the cylinder's longitudinal axis and line OP at the edge of the step (as illustrated in the diagram). Determine the minimal magnitude of the force F required to surmount the cylindrical body over the step. (Take $g = 10\ m/s^2$)



(1) $50\sqrt{3}$ N (2) $25\sqrt{2}$ N

(3) $25\sqrt{3}$ N (4) $50\sqrt{2}$ N

16. In the figure shown, objects, P, Q, and R, are arranged in a linear configuration upon a frictionless horizontal surface. Objects P and Q each possess a mass of m, while object R has a mass of 2 m. Object P moving with a velocity of 10 m/s approaches Object Q, collides completely inelastically. Subsequently, object Q then collides with object R resulting in a perfectly elastic collision. All motions occur along the same straight line. Determine the final velocity of object R.



(1) 4.5 m/s (2) 9 m/s

(3) 2.25 m/s (4) 5 m/s

17. A particle of mass 2 kilograms exhibits a time-dependent velocity vector expressed as: $\vec{v}(t) = \left(\sqrt{t}\hat{i} + t^3\hat{j}\right)\ ms^{-1}$ where $\vec{v}(t)$ is in meters per second and t is in seconds. Determine the magnitude of the net force acting upon the particle at the instant t = 1 second.

(1) 37 N (2) $\sqrt{37}$ N

(3) 36 N (4) $\sqrt{39}$ N

18. Two athletes engage in a race along a straight path. Athlete A completes the race 10 seconds sooner than athlete B, crossing the finish line with a velocity that exceeds that of athlete B by a magnitude of v. Assuming both athletes commence from rest and maintain constant accelerations of 4 m/s^2 and 2 m/s^2 respectively, determine the value of v

(1) $\frac{80}{3}$ m/s (2) 40 m/s

(3) $20\sqrt{2}$ m/s (4) 30 m/s

19. Consider a mass immersed in a gravitational field produced by a spherically symmetric mass distribution characterized by a density function $\rho(r) = K/r$, where r denotes the radial distance from the center of the mass distribution. The mass is executing a circular orbit of radius R within this gravitational field. The correct relation relating the radius R of the particle's orbit to its orbital period T is

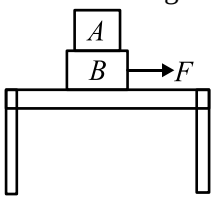
(1) $\frac{T^2}{R}$ = constant

(2) $\frac{T}{R}$ = constant

(3) T^2R = constant

(4) TR^2 = constant

20. Two rectangular objects, A and B, are positioned upon a horizontal surface, as illustrated in the diagram. Object A possesses a mass of 2 kilograms, while object B has a mass of 3 kilograms. The coefficient of static friction between objects A and B is 0.6, and the coefficient of static friction between object B and the surface of the table is 0.4. Determine the maximum horizontal force F that can be applied to object B without causing object A to slide relative to object B. The acceleration due to gravity is 10 m/s^2.

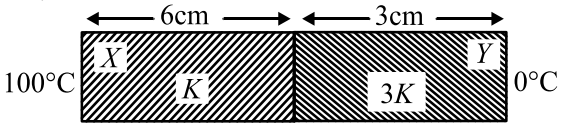


(1) 20 N (2) 50 N

(3) 30 N (4) 60 N

Integer Type Questions

21. Two planar objects, X and Y, are joined together to form a composite material. Object X possesses a thermal conductivity of K and a thickness of 6 cm, while object Y exhibits a thermal conductivity of 3K and a thickness of 3 cm. The cross-sectional area of each object is 150 cm^2. Determine the value of α in the following expression, if the equivalent thermal conductivity is $K_{eq} = (1 + 2/\alpha)K$.



22. A scientist conducts an experiment to understand the behaviour of gas. In this experiment, a heat of 100 J is provided to the gaseous system which then undergoes an isobaric expansion. The work done by the gas (in J) during this process is (For the given molecule, degree of freedom is 6)

23. A ring of mass 6 kg and radius 10cm is free to rotate about a fixed horizontal axis passing through its centre and perpendicular to the plane of ring. A point object of mass 2 kg is fixed at the highest point of the ring. Now the system is released. When the body comes to the lowest position, its angular speed (in $rads^{-1}$) will be (Take $g = 10 m/s^2$)

24. A physicist is studying the behaviour of a simple harmonic oscillator. He observes the oscillator's motion, noting that its displacement from equilibrium is described by the equation $x(t) = A\sin(\omega t + \varphi)$ where A is the amplitude, ω is the angular frequency. At time $t = 0$ seconds, the physicist measures the oscillator's position to be 3 cm and its velocity to be ω cm/s. To analyse the oscillator's motion further, the physicist needs to determine its amplitude. The amplitude of the harmonic oscillator is found to be $\sqrt{y}$ cm. The value of y is

25. Two ships are in motion. Ship A is moving along the East with its position at an instant t given by $s_1(t) = 10 - 3t^2$ in eastward direction. Ship B is moving along the North with its position at time t given by $s_2(t) = 5 - 6t + 8t^3$ in northward direction. At the instant $t = 1$ second, ship B appears to be moving at a speed of x m/s relative to the ship A. The value of x^2 is

SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. Consider the following statements about the modern periodic table and identify the correct ones among them:

(A) Elements in the same group have similar chemical properties due to the same number of valence electrons.

(B) The atomic radius generally increases as you move down a group.

(C) Elements in the same period have similar chemical properties.

(D) The first ionization energy generally increases across a period from left to right.

(1) A and B only (2) A, B, and D only

(3) A, C, and D only (4) A, B, C, and D

27. When the temperature of water increases, its pH typically decreases. Which of the following statements accurately explains this observation?

(1) The increase in temperature leads to a decrease in the concentration of H^+ ions in the water..

(2) The dissociation of water into H^+ and OH^- ions absorbs heat, making the solution less acidic.

(3) The increase in temperature increases the dissociation of water into H^+ and OH^- ions, thereby increasing the concentration of H^+ ions and decreasing the pH.

(4) Higher temperatures decrease the solubility of H^+ ions in water, which lowers the pH.

28. An example of disproportionation reaction is:

(1) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(2) $H_2S + 3O_2 \rightarrow 2H_2SO_4$

(3) $2CuI_2 \rightarrow 2CuI + I_2$

(4) $2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O$

29. Which of the following pairs includes one molecule with an odd number of electrons and another molecule with an incomplete octet?

(1) NO_2 and BF_3

(2) O_3 and CO_2

(3) ClO_3^- and NH_3

(4) NO and H_2O_2

30. A sample of 15.0 g of a solid compound X(M.Wt. of X is 75 g/mol) is placed in a 5.0 L container at 300°C. If 25% of X decomposes into two gaseous products, Y and Z $\left(\underset{(s)}{x} \rightleftharpoons \underset{(g)}{y} + \underset{(g)}{z}\right)$. At the same temperature, the equilibrium constant Kp for the reaction is :

(1) $0.242 \times 10^{-4}\ atm^2$

(2) $0.2213\ atm^2$

(3) $4.9 \times 10^{-3}\ atm^2$

(4) $0.342\ atm^2$

31. At 298 K and 1 atm pressure, 15 mL of an unknown hydrocarbon gas requires 75 mL of oxygen (O_2) for complete combustion. The reaction produces 60 mL of carbon dioxide (CO_2). What is the molecular formula of the hydrocarbon?

(1) C_3H_6 (2) C_5H_{10}

(3) C_4H_4 (4) C_2H_4

32. The region in the electromagnetic spectrum where the Paschen series lines appear is:

(1) Ultraviolet (2) Visible

(3) Infrared (4) Microwave

33. For the following Assertion and Reason, the correct option is:

Assertion (A): Aluminium is typically found in the +3 oxidation state, while thallium can exist in both +1 and +3 oxidation states.

Reason (R): Group 13 elements down the group show inert pair effect.

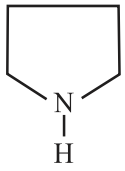
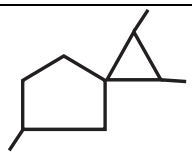
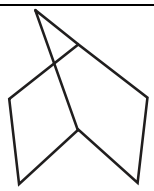
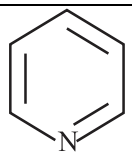
(1) Assertion is true, but reason is false.

(2) Both assertion and reason are false.

(3) Both assertion and reason are true but the reason is not the correct explanation for the assertion

(4) Both assertion and reason are true, and the reason is the correct explanation for the assertion

34. Match the compounds in column-I with their types in column -II.

Column-I		Column-II	
A		P	Follows $(4n+2)\pi e^-$ Rule
B		Q	Non-planar Heterocyclic compound
C		R	Compound with two bridgehead 'C' atoms
D		S	Cyclic compound containing quarternary carbon atoms

Choose the correct answer from the options given below:

	A	B	C	D
(1)	Q	S	P	R
(2)	P	R	S	Q
(3)	Q	S	R	P
(4)	S	R	P	Q

35. Consider the following reaction

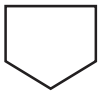
$$CH_3(CH_2)_3CH_3 \xrightarrow[HCl,\Delta]{\text{Anhy.}AlCl_3} \underset{\text{Major Product}}{'X'}$$

X is:

(1) $CH_3(CH_2)_3CH_2Cl$

(2) $Cl-CH_2-(CH_2)_3-CH_2-Cl$

(3) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(4) 

36. Which of the following compounds is distorted octahedral in shape:

(1) IF_5 (2) XeF_6

(3) SF_6 (4) ClF_5

37. Select the pair with correct order of bond enthalpy ($kJmol^{-1}$)

(1) C – C > Si – Si and N – N > P – P

(2) N – N > P – P and O – O > S – S

(3) C – C > Si – Si and O – O < S – S

(4) N – N > P – P and O – O < S – S

38. Which technique is commonly employed to isolate ethanol from a mixture in the beverage industry?

(1) Column chromatography

(2) Fractional distillation

(3) Sublimation

(4) Recrystallization

39. When 10 moles of a certain acid, are neutralized by 15 moles of NaOH, it is found that the acid is a diprotic acid. What is the number of moles of H_2X used in the reaction? (H_2X is a weak acid)

(1) 20 moles

(2) 10 moles

(3) 15 moles

(4) 7.5 moles

40. Given the Thomson model of the atom, which of the following observations would be expected in an experiment where α-particles are directed at a thin metal foil?

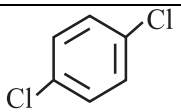
(1) Large number of α-particles are scattered at very large angles due to a dense nucleus.

(2) α-particles pass through the foil with minimal deflection, indicating the atom is mostly empty space.

(3) α-particles are completely absorbed by the foil with no particles passing through.

(4) α-particles experience significant energy loss while passing through the foil.

41. Match column-I with column -II.

Column-I		Column-II	
A		P	B.O. = 2.5
B	O_2^-	Q	$\mu \neq 0$
C		R	B.O. = 1.5
D	O_2^+	S	$\mu = 0$

Choose the correct answer from the options given below column

	A	B	C	D
(1)	Q	P	S	R
(2)	S	R	Q	P
(3)	Q	R	S	P
(4)	S	P	Q	R

42. For a monatomic ideal gas in a closed system, which of the following statements is true regarding the thermodynamic properties of the gas?

(1) The internal energy U is independent of temperature.

(2) The specific heat at constant volume C_V increases with an increase in volume.

(3) The ratio of specific heats $\gamma = C_P/C_V$ is equal to 1.

(4) The specific heat at constant pressure C_P is greater than the specific heat at constant volume C_V.

43. At 298 K, the solubility of calcium phosphate $Ca_3(PO_4)_2$ in water is 1.0×10^{-7} mol L^{-1}. Calculate the concentration of phosphate ions $[PO_4^{3-}]$ in a saturated solution of calcium phosphate.

(1) 3.0×10^{-7} mol L^{-1}

(2) 2.0×10^{-7} mol L^{-1}

(3) 1.0×10^{-7} mol L^{-1}

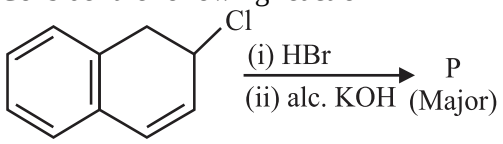
(4) 6.0×10^{-7} mol L^{-1}

44. Which of the following compounds can only act as an oxidizing agent and not as a reducing agent?

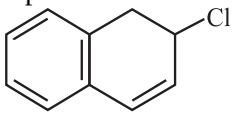
(1) HNO_3 (2) H_3PO_3

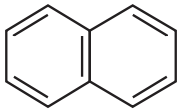
(3) H_2SO_3 (4) H_2O_2

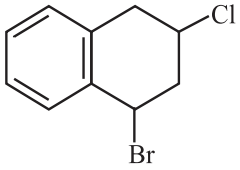
45. Consider the following reaction

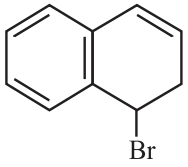


Major product P is:

(1) 

(2) 

(3) 

(4) 

Integer Type Questions

46. How many of the following properties are intensive :

Density, Heat capacity, Molarity, Entropy, Specific volume, Molar mass, Boiling point, Refractive index

47. The atomic number of the element ununquadium is x____ The value of $\frac{x}{10}$ is (nearest integer)

48. At a constant temperature and volume, a gaseous reactant A decomposes according to the reaction:

$A(g) \rightleftharpoons 2B(g)+C(g)$

Initially, the pressure of A is 500 mm Hg. After a certain amount of time, the total pressure in the container is observed to be 800 mm Hg. At the constant volume of the container and constant temperature, if the fraction of A remaining is "a" then the value of "10a" is:

49. A sample of an organic compound weighing 1.50 g is analyzed using Duma's method. The nitrogen gas collected after combustion occupies a volume of 320.0 mL at a temperature of 30.0°C and a pressure of 1.2 atm. Calculate the percentage of nitrogen in the sample.(Given: Universal Gas constant R: 0.0821 L atm mol^{-1} K^{-1} (Answer to the nearest integer value.)

50. $CH_3-CH_2-C\equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$

$$(A)\xrightarrow[\text{Anhyd.}AlCl_3]{CH_3-\overset{\overset{O}{\|}}{C}-Cl\ (1\ eq)}(B)$$

No. of primary hydrogen present is product (B) is/are_____

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. The number of five-digit numbers, greater than 40004 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to

(1) 120 (2) 132

(3) 72 (4) 96

52. Let $A=\left\{x\in\mathbb{R}:\left[x+3\right]+\left[x+4\right]\le 5\right\}$

$B=\left\{x\in\mathbb{R}:3^{x}\left(\sum_{r=1}^{\infty}\frac{3}{10^{r}}\right)^{x-3}<3^{-3x}\right\}$, where $[t]$ denotes greatest integer function. Then,

(1) $A\cap B=\phi$

(2) $A=B$

(3) $B\subset A, A\neq B$

(4) $A\subset B, A\neq B$

53. A group of students comprises of 6 boys and n girls. A team of 3 students is to be formed from randomly such that there is at least one boy and at least one girl in each team, is 1440, then n is equal to:

(1) 24 (2) 28

(3) 20 (4) 26

54. Let $f(x) = ax^2 + bx + c$ where $a, b, c \in R$. be a quadratic polynomial such that $f(2) + f(3) = 0$. If one of the roots of $ax^2 + bx + c = 0$ is –1, then the sum of the roots of $ax^2 + bx + c = 0$ lies in interval

(1) (–2, –1) (2) (–1, 1)

(3) (1, 2) (4) (2, 3)

55. Let x_i; y_i; (for i = 1, 2, 3, ….. n) be positive real numbers such that $\sum_{i=1}^{n}x_i=\sum_{i=1}^{n}y_i=9$. If m be the minimum possible value of $\log_3\left(\sum_{i=1}^{n}3^{y_i}\right)$ and M be the maximum possible value of $\left(\sum_{i=1}^{n}\log_3 x_i\right)$, then $\lim_{n\to\infty}\left(m-\log_3 n\right)\left(M+n\log_3 n\right)$ is.

(1) 3

(2) 18

(3) 3/2

(4) 9/2

56. Let α, β be the roots of the quadratic equation $x^2+\sqrt{6}x+3=0$. Then $\left[\frac{\alpha^{24}+\alpha\beta^{23}+\beta\alpha^{23}+\beta^{24}}{\alpha^{23}+\beta^{23}}\right]^2$ is equal to

(1) 79 (2) 72

(3) 6 (4) 9

57. Let x_1, x_2 …., x_{100} be in an arithmetic progression, with x_1 = 2 and their mean equal to 200. If $y_i = i(x_i - i), 1 \le i \le 100$, then the mean of y_1, y_2, …., y_{100} is

(1) 10101.50

(2) 10051.50

(3) 10049.50

(4) 10100

58. Let $a_1 = 2, a_2, a_3, \ldots a_n$ be a G.P. If the product of its first four terms is 11664 and the product of its last four terms is 16×3^{70}, then the product of its middle two terms is

(1) 2×3^{19} (2) 4×3^{19}

(3) 2×3^{18} (4) 4×3^{18}

59. The complex number z_1 and z_2 are such that they are neither purely real nor purely imaginary. If $Im(z_1 z_2) = 0$ and $Im(z_1 + z_2) = 0$, then which of the following are possible?

A. $Re\,(z_1) > 0$ and $Re\,(z_2) > 0$

B. $Re\,(z_1) < 0$ and $Re\,(z_2) > 0$

C. $Re\,(z_1) > 0$ and $Re\,(z_2) < 0$

D. $Re\,(z_1) < 0$ and $Re\,(z_2) < 0$

Choose the correct answer from the options given below:

(1) *A* and *D*

(2) *B* and *C*

(3) *A* and *B*

(4) *A* and *C*

60. Let $A=\left\{\theta\in[0,2\pi]:\frac{3+2i\sin\theta}{4-i\sin\theta}\text{ is purely real}\right\}$

Then the sum of the elements in A is

(1) π

(2) 2 π

(3) 4 π

(4) 3 π

61. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{l}}\right)^{9}$ is –84 and the coefficient of x^{-3l} is $2^{\alpha}\beta$, where β is an odd negative integer. Then α is equal to

(1) 6 (2) 5

(3) 7 (4) 3

62. Let $x=\left(3+2\sqrt{2}\right)^{17}$ and if $[t]$ denotes the greatest integer $\le t$, then

(1) x is an even integer

(2) x is an odd integer

(3) $[x]$ is an even integer

(4) $[x]$ is an odd integer

63. Let the sample space $S=\left\{x\in Z: x(70-x)\ge 24M\right\}$ where M is the maximum value of product of two positive integer whose sum is 10 and the event $A=\{x\in S: x$ is not a multiple of 3$\}$. Then $P(A)$ is equal to

(1) $\frac{16}{49}$ (2) $\frac{1}{3}$

(3) $\frac{2}{3}$ (4) $\frac{17}{49}$

64. The value of $\lim\limits_{x\to 0}\left(\frac{\sin x+\sin 3x}{3\sin x-\sin 3x}\right)\cdot x^2$ is

(1) 1 (2) 2

(3) 3 (4) 4

65. The value of

$$64\sin\left(\frac{31\pi}{66}\right)\cdot\sin\left(\frac{29\pi}{66}\right)\sin\left(\frac{25\pi}{66}\right)\cdot\sin\left(\frac{17\pi}{66}\right)\sin\left(\frac{\pi}{66}\right)$$

is equal to

(1) 3 (2) 2

(3) 4 (4) 1

66. If $0<x,y<\pi$. and $\cos x+\cos y-\cos(x+y)=3/2$, Then, minimum value of $\cos\frac{x}{2}\cdot\cos\frac{y}{2}$ is equal to:

(1) $\frac{1+\sqrt{3}}{2}$ (2) $\frac{1}{2}$

(3) $\frac{1}{8}$ (4) $\frac{1}{\sqrt{2}}$

67. The domain of $f(x)=\frac{\log_{(x+1)}(x+2)}{e^{2\log_e x}-(2x+3)}$ is

(1) $\mathbb{R}-\{1,3\}$ (2) $(2,\infty)-\{3\}$

(3) $(0,\infty)-\{3\}$ (4) $\mathbb{R}-\{3\}$

68. The chord $AB = ax + by = 0$, $(a \ne b)$ of the circle $x^2 + y^2 - 2x = 0$ where $A \equiv (\alpha, 0)$ and $B \equiv (1, \beta)$, where $\beta > 0$, is taken as diameter to form a circle C. The image of the circle C in the line $y = -x$ is:

(1) $x^2+y^2+x+y=0$

(2) $x^2+y^2+x-y=0$

(3) $x^2+y^2+x+y+\frac{3}{2}=0$

(4) $x^2+y^2-x+y+\frac{3}{2}=0$

69. A circle in the first quadrant touches the two coordinate axes at the points A and B. From a point P (α, β) lying on the circle above the line AB, Perpendicular is drawn on AB. If length of this perpendicular is equal to 12 units, then the value of $\alpha\beta$ is_____

(1) 144

(2) 142

(3) 140

(4) 145

70. The radius of the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$ is $\frac{a}{\sqrt{5}}$. The value of a is

(1) 6

(2) 5

(3) 8

(4) 7

Integer Type Questions

71. Let $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1, a>0, b>0$, be a hyperbola such that the length of it's transverse axis is $8\sqrt{2}$. If the eccentricity is $\frac{\sqrt{11}}{2}$, then the length of its conjugate axis is $4\sqrt{k}$. The value of k, is

72. Let $|z_1|= 3$ and $|z_2 - (3 + 4i)| = 2$ then maximum value of $|z_1 - z_2|$ is

73. Set A and B have m and n element such that sum of their number of subsets is 288 then $|m - n|$ is

74. The mean age of all the teachers in a school is 42 years. A teacher retires at the age of 60 years and a new teacher aged 35 years is appointed in his place. If the mean age of the teachers in this school now is 41 years, then the number of teachers in the school is

75. The number of 5 digit number formed by exactly two digits is 9k. The value of k is

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